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Selective Inference for Hierarchical Clustering

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1 Introduction

Clustering algorithms enable unsupervised partitioning of data. By identifying hidden structures without requiring strong prior assumptions, they have found widespread application across diverse domains. However, in general, clustering methods do not provide theoretical guarantees regarding the partitions they generate in standard scenarios. To ensure that the results provide guarantees on the actual differences between groups, a pertinent strategy would be to perform tests for equality of means between the different clusters.

However, implementing a standard z-test or Wald test results in an inflation of the type I error rate [5]. This arises from the fact that the null hypothesis is constructed adaptively, that is, it is defined only after observing the data. This issue is analogous to overfitting in supervised learning frameworks. When the same data set is used for both model training and evaluation, the resulting model often fails to generalize, capturing noise rather than underlying patterns. In the context of inference, the use of the same data for model selection and inference is referred as *double-dipping*. In post-clustering inference, the null hypothesis is data-driven and classical inference prevents type I error control. Note that simply splitting the dataset into a training set and a test set does not resolve the issue. Indeed, even if clusters are estimated using the training data and mean equality tests are performed on the testing data, one must first assign each test point to a cluster. Consequently, this assignment necessarily reuses the testing data for both predicting a clustering and testing the null hypothesis.

To address double-dipping, Gao *et al.* developed a valid test for the difference in means between two clusters estimated via an hierarchical agglomerative clustering algorithm. Their method assumes that the observations are independent and identically distributed as p-dimensional Gaussians with spherical covariance. This corresponds to considering the following matrix normal model:

$$\mathbf{X} \sim \mathcal{MN}_{n\times p}(\boldsymbol{\mu}, \mathbb{I}_n, \sigma^2\mathbb{I}_p)$$

where $\boldsymbol{\mu} \in \mathbb{R}^{n\times p}$ and $\sigma > 0$. The authors in [5] defined a p-value that controls the so-called selective type I error, and that can be efficiently computed numerically for hierarchical agglomerative clustering algorithms.

Here, we introduce the post-clustering inference framework presented in [5]. This work is organized as follows:

- In Section 2, we revisit the hypothesis testing framework from a more general perspective. Then, we explore examples that motivate selective inference.

- In Section 3.1, we present the approach of Gao *et al.* to address selective inference in the context of clustering. Then, in Section 3.2, we

illustrate the validity of the method and evaluate its robustness to feature dependence. Finally, in Section 3.3 we assess the suitability of the method when applied to real genotype data.

For clarity and readability, all technical proofs have been relegated to Appendix Appendix A.

2 Classical versus selective testing

2.1 Hypothesis testing and *p-values*

In this section, we rigorously establish the framework for hypothsis testing. Let us first recall some basic concepts of probability theory. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, $(E, \mathcal{E})$ a topological space and $\mathcal{T}$ the σ-algebra generated by $\mathcal{E}$. A *random variable* is a measurable function $X : (\Omega, \mathcal{F}) \to (E, \mathcal{T})$. The *(probability) distribution* of X is the mapping $P : \mathcal{T} \to [0, 1]$ such that $P(O) = (\mathbb{P} \circ X^{-1})(O)$ for all $O \in \mathcal{T}$. We say that P is *supported* on E and denote by $\mathcal{M}_1(E)$ the set of all probability distributions supported on $\mathcal{E}$. From now on, we will set $E = \mathbb{R}$ and simply write $\mathcal{M}(\mathbb{R}) = \mathcal{M}$.

Definition 1 (Hypothesis, Definition 1.1 in [7])**.** *A hypothesis $\mathcal{H}_0$ is a set of probability distributions in $\mathcal{M}$. A hypothesis is simple if it is a singleton, such as $\{P\}$ or $\{Q\}$, and composite otherwise. The complementary set $\mathcal{M} \setminus \mathcal{H}_0$ is called the alternative hypothesis of $\mathcal{H}_0$.*

Definition 2 (Test)**.** *A test for $\mathcal{H}_0$ is a binary partition of the sample space, defined by a mapping:*

$$\begin{array}{rcl} \pi_{\mathcal{H}_0} : \Omega & \to & \{0, 1\} \\ \omega & \mapsto & \pi_{\mathcal{H}_0}(\omega). \end{array} \tag{1}$$

For $\omega \in \Omega$, we say that the test rejects $\mathcal{H}_0$ based on ω if $\pi_{\mathcal{H}_0}(\omega) = 1$, and does not reject $\mathcal{H}_0$ based on ω if $\pi_{\mathcal{H}_0}(\omega) = 0$.

In what follows, we will adopt the more standard notation:

$$\mathbb{P}\left(\text{Reject } \mathcal{H}_0\right) = \mathbb{P}\big(\{\omega \in \Omega \,:\, \pi_{\mathcal{H}_0}(\omega) = 1\}\big),$$

making the dependence on $\pi_{\mathcal{H}_0}$ implicit. In order to make informative and meaningful decisions, a test must ensure that rejection is stated *with stastical guarantees*. More precisely, we say that a test is valid if it controls the type I error, defined as follows.

Definition 3 (Type I error)**.** *We say that a test controls the type I error at level α if*

$$\mathbb{P}(\text{Reject } \mathcal{H}_0) \leq \alpha, \quad \textit{for } \alpha \in (0, 1).$$

We say that it controls the type I error exactly at level α if

$$\mathbb{P}(\text{Reject } \mathcal{H}_0) = \alpha, \quad \textit{for } \alpha \in (0, 1).$$

Type I error control can be characterized through a specific family of random variables, known as *p-values*. To properly define this quantity, let us first equip the space of random variables with the following partial order relation.

Definition 4 (Stochastic dominance)**.** *Let X and Y be real-valued random variables. We say that Y stochastically dominates X, and write*

$$X \preceq_{\text{st}} Y,$$

if

$$\mathbb{P}(X \leq u) \geq \mathbb{P}(Y \leq u), \quad \forall\, u \in \mathbb{R}. \tag{2}$$

A case of particular interest is the stochastic dominance of uniform random variables, which leads to the definition of the *super-uniform* family of distributions.

Definition 5 (Super-uniform random variable)**.** *Let X be a real-valued random variable. We say that X is super-uniform on [0,1], and write $X \sim \mathcal{SU}(0,1)$, if X stochastically dominates a uniform random variable on $[0,1]$, that is, if*

$$\mathbb{P}(X \leq u) \leq u, \quad \forall\, u \in [0,1]. \tag{3}$$

A super-uniform random variable can be recovered from any pair of ordered random variables.

Proposition 1. *Let X and Y be real-valued random variables, with cumulative distribution functions F_X and F_Y, respectively. If $X \preceq_{\text{st}} Y$, then $F_X(Y) \sim \mathcal{SU}(0,1)$.*

Remark 1. *If $X = Y$, then $F_X(X)$ is super-uniform on $[0,1]$. Moreover, if X has a continuous distribution function F_X, then*

$$F_X(X) \sim \mathcal{U}(0,1). \tag{4}$$

The notion of super-uniformity motivates a natural definition of p-value.

Definition 6 (p-value, Definition 1.1 in [7])**.** *Let $\mathcal{H}_0$ be a hypothesis. A p-value for $\mathcal{H}_0$ is a super-uniform random variable under $\mathcal{H}_0$.*

Indeed, a p-value can be defined as a random variable supported on $[0,1]$ that is super-uniform under the null hypothesis. This justifies the usual construction of tests of the form $\pi_{\mathcal{H}_0} = \mathbb{1}\{p \leq \alpha\}$.

Proposition 2. *Let $\mathcal{H}_0$ be a hypothesis and p a p-value for $\mathcal{H}_0$. Then, the test $\pi_{\mathcal{H}_0} = \mathbb{1}\{p \leq \alpha\}$ controls the type I error at level α, for $\alpha \in (0,1)$.*

We usually define p-values as transformations of *test statistics*, defined below.

Definition 7 (Test statistic)**.** *A test statistic is a measurable function* $T : \Omega \to \mathbb{R}$*.*

From now on, we will consider the case of unilateral tests. In this setting, to define a test based on the information given by a real-valued random variable X, p-values are often built in the form:

$$p(x) = \mathbb{P}_{\mathcal{H}_0}\big(T(X) \geq T(x)\big), \tag{5}$$

where T is a test statistic and x a realization of X.

It is straightforward to derive that the p-value (5) follows a uniform distribution under $\mathcal{H}_0$. However, we have defined p-values as being super-uniform random variables under the null hypothesis, which corresponds to a more general family of distributions. To adapt the classical form (5) to that setting, we introduce the following proposition.

Proposition 3. *Let T and T' be two test statistics. Let X be a random variable and x a realization of X and consider*

$$p(x) = \mathbb{P}_{\mathcal{H}_0}\big(T'(X) \geq T(x)\big). \tag{6}$$

Then, if

$$T'(X) \preceq_{\text{st}} T(X) \quad \textit{under } \mathcal{H}_0, \tag{7}$$

the quantity p is a p-value for $\mathcal{H}_0$.

Remark 2. *If $T' = T$ and the cumulative distribution function F_T of $T(X)$ is continuous under $\mathcal{H}_0$, we have*

$$p \overset{\mathcal{H}_0}{\sim} \mathcal{U}(0,1).$$

The proof of all the results presented in this section are presented in Appendix A.1. In the classical setting, the null hypothesis is independent of the data. However, in many practical applications $\mathcal{H}_0$ is chosen *after seeing the data*. In this framework, the classical testing approaches built for type I error control are unsuitable. Instead, statistical guarantees need to be provided via *selective inference*. In particular, the theory of statistical testing of data-driven null hypotheses is known as *selective testing* [4]. The next section provides some examples that motivate this framework.

2.2 Examples motivating selective testing

2.2.1 Lasso

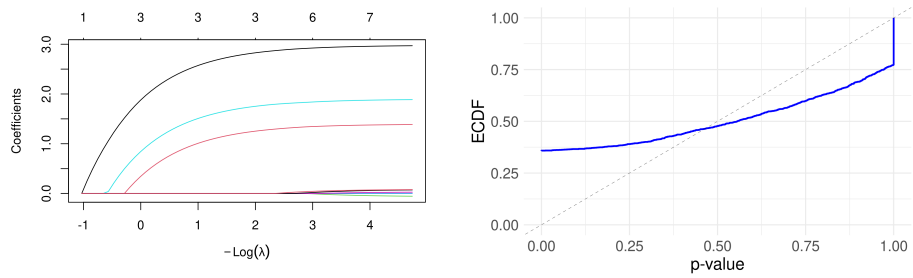
To determine whether an explanatory variable helps explain a response variable through a linear regression model, a common practice is to test whether its associated coefficient is significantly different from zero. In the context of classical linear regression, this falls within the framework of non-selective

inference, because all coeficients are fixed *a priori*. In contrast, Lasso regression uses an ℓ_1-penalty to perform variable selection [8]. As only the coefficients selected by the Lasso can be tested, the null hypothesis depends on the outcome of the Lasso regression, and is therefore data-driven. This corresponds to a setting of selective inference, where the classical control of type I error fails.

To illustrate the unsuitability of non-selective inference, consider a centered Gaussian vector $X = (X_1, \ldots, X_8)$ and set the model

$$Y = X\beta + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0,1), \quad \beta = (3, 1.5, 0, 0, 2, 0, 0, 0)^\top.$$

We generated $n = 100$ independent realizations of (X, Y) and implemented the Lasso algorithm, obtaining the regularization path presented in Figure 1(a). Then, we tested whether a randomly chosen coefficient selected by Lasso regression and among the trully null coefficients, equals 0. After repeating this pipeline $M = 2000$ times, we obtained the empirical p-value distribution depicted in Figure 1(b). We clearly see that the p-values are not super-uniform (recall Definition 5), motivating the use of a selective testing approach instead of naive testing after selection.



(a) (b)

Figure 1: (a) Regularization path of the coefficients obtained using the Lasso algorithm with a single realization of (X, Y). Some coefficients are zero, and three converge toward the theoretical values (3,1.5,2). (b) Empirical cumulative distribution function (ECDF) of the p-values obtained from tesing wheter a randomly chosen coefficient is null after selecting the coefficient by a Lasso regression. The ECDF was computed from $M = 2000$ simulations.

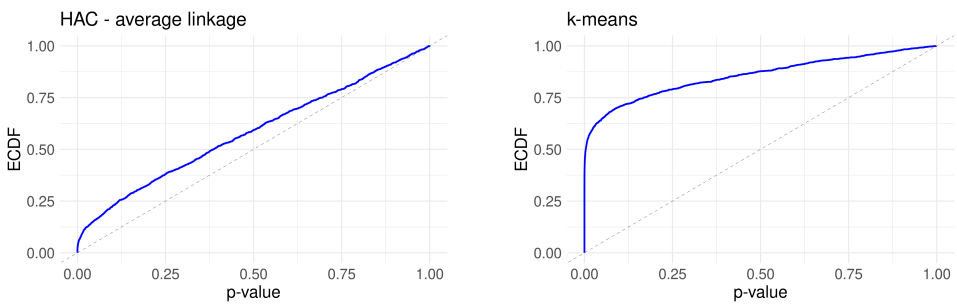
2.2.2 Publication bias

In many areas of research, scientists tend to test for signficance only when the associated effect is found to be substantial. In other words, testing is performed after a selection process that filters out small effects. In that setting, controlling type I error at a fixed level α yields inflated false positive

rates when considering the ensemble of all published studies. If $Y_i \sim \mathcal{N}(\mu_i, 1)$ represents the effect size in a scientific study, whose significance is tested only if $|Y_i| > 1$, a naive level α test $H_{0,i} : \mu_i = 0$ is invalid. Indeed, Fithian [4] demonstrates that the false positive rate among true nulls reaches approximately 0.16, far exceeding the nominal 0.05 level. Valid inference requires thresholding $|Y_i|$ at 2.41 rather than 1.96, the 0.95 quantile of the standard normal, imposing a more stringent criterion.

2.2.3 Clustering

Another remarkable example of selective inference appears when evaluating the performance of clustering algorithms by testing for the equality of cluster means. To illustrate the need of using appropriate tests in the context of selective inference, we simulated $n = 100$ samples of a five-dimensional standard Gaussian random vector. Each sample was classified into $K = 3$ groups using hierarchical agglomerative clustering and k-means. Then, the equality of cluster means was tested using a classical t-test, for two randomly selected clusters. If the test controlled the type I error in this setting, the resulting p-value would be super-uniformly distributed under the null hypothesis. However, the 'clustering + post-selection testing' procedure leads to a deviation from super-uniformity, as shown in Figure 2.



(a) (b)

Figure 2: Empirical cumulative distribution functions of the p-values obtained after testing the equality of cluster means for (a) hierarchical agglomerative clustering (HAC) and (b) k-means algorithms. The distribution functions were computed using $M = 2000$ simulations from a multivariate centered normal distribution.

2.3 Addressing selective testing

Since the type I error is no longer controlled, the definition of alternative error risks is required in this setting. Accordingly, we present three methods as described by Yoav Benjamini [2].

2.3.1 Simultaneous Inference

This approach controls the family-wise error rate across all hypotheses:

$$\mathbb{P}(\text{At least 1 false positive among all hypotheses}) \leq \alpha.$$

This strategy proves highly conservative, ensuring that for every possible set of hypotheses, the probability of at least one false positive remains below α. To date, no post-clustering inference approaches based on this type of control have been proposed.

2.3.2 Sample splitting

This approach consists in splitting the dataset into a training set X and a test set Y. The training set X is used to choose the null hypothesis $\mathcal{H}_0(X)$. Then, inference is performed on the test set Y. Although relatively simple to implement, this method raises several issues. First, statistical guarantees on the tests hold only if X and Y are independent, which is not straightforward in practice. In addition, comparing cluster means on the observations in Y requires assigning each test point to one of the clusters obtained from the clustering performed on X, a step that compromises validity, as it reuses the dataset Y for two separate steps (predicting a clustering and performing the inference). As discussed in [5], this strategy does not yield valid post-clustering inference in general.

2.3.3 Conditional Inference

This approach constitutes the most extensively studied framework for post-clustering inference. It controls the false positive probability conditional on hypothesis selection:

$$\mathbb{P}(\text{Reject } \mathcal{H}_0(X) \text{ being true} \mid \mathcal{H}_0 \text{ selected}) \leq \alpha. \tag{8}$$

Conditional inference largely dominates the state-of-the-art, being the basis of the approach proposed in [5]. In the remainder of the article, we employ this method to develop a statistical procedure for testing equality of cluster means following a clustering algorithm.

3 Post-clustering inference

The article under study [5] was the first one to propose a feasible solution to post-clustering inference. Here, we will focus on the setting and approach introduced by the authors.

3.1 Gao *et al.*'s approach

Let $\mathbf{X} = (X_{ij})_{ij} \in \mathbb{R}^{n \times p}$ be the design matrix. A cluster is an element of a partition of the samples. We note C a clustering algorithm and $C_1 \in C(\mathbf{X})$ a cluster obtained by the algorithm C on $\mathbf{X}$. Let $\boldsymbol{\mu} = (\mu_{ij})_{ij}$ be the mean matrix of $\mathbf{X}$, that is, $\mu_{ij} = \mathbb{E}[X_{ij}]$ and, for a subset G of $\{1, \ldots, n\}$, we write

$$\bar{\mu}_G = \frac{1}{|G|} \sum_{i \in G} \mu_i \in \mathbb{R}^p \quad \text{and} \quad \bar{X}_G = \frac{1}{|G|} \sum_{i \in G} X_i \in \mathbb{R}^p.$$

The goal is to test for the equality of cluster means, that is, testing the following null hypothesis:

$$\mathcal{H}_0^{\{C_1, C_2\}} : \bar{\mu}_{C_1} = \bar{\mu}_{C_2}, \tag{9}$$

for a pair of clusters $C_1, C_2 \in C(\mathbf{X})$. For concision and clarity, we introduce the n-dimensional vector

$$\nu(C_1, C_2) = \nu = (\nu_i)_{i=1}^n, \qquad \nu_i = \begin{cases} \dfrac{1}{|C_1|}, & \text{if } i \in C_1, \\ -\dfrac{1}{|C_2|}, & \text{if } i \in C_2, \\ 0, & \text{otherwise,} \end{cases} \tag{10}$$

such that the population (resp. empirical) difference of cluster means can be compactely rewritten as:

$$\begin{cases} \bar{\mu}_{C_1} - \bar{\mu}_{C_2} = \nu^T \boldsymbol{\mu} \\ \bar{X}_{C_1} - \bar{X}_{C_2} = \nu^T \mathbf{X}. \end{cases}$$

Thus, we can rewrite the null hypothesis as follows:

$$\mathcal{H}_0^{\{C_1, C_2\}} : \bar{\mu}_{C_1} = \bar{\mu}_{C_2} \iff \nu^T \boldsymbol{\mu} = 0. \tag{11}$$

From now on, we adopt the conditional inference framework, described in Section 2.3.3. In this setting, controlling the risk (8) is equivalent to controlling the so-called *selective type I error for clustering*, defined below.

Definition 8 (Selective type I error for clustering)**.** *We say that a test controls the selective type I error for clustering at leval α if*

$$\mathbb{P}_{\mathcal{H}_0^{\{C_1, C_2\}}}\big(\textit{Reject } \mathcal{H}_0^{\{C_1, C_2\}} \mid C_1, C_2 \in C(\mathbf{X})\big) \leq \alpha, \quad \alpha \in (0, 1). \tag{12}$$

We say that it controls exactly the selective type I error for clustering at level α if

$$\mathbb{P}_{\mathcal{H}_0^{\{C_1, C_2\}}}\big(\textit{Reject } \mathcal{H}_0^{\{C_1, C_2\}} \mid C_1, C_2 \in C(\mathbf{X})\big) = \alpha, \quad \alpha \in (0, 1). \tag{13}$$

To define a p-value that controls the selective type I error, model assumptions need to be imposed on $\mathbf{X}$. The authors in [5] adopt the following matrix normal model:

$$\mathbf{X} \sim \mathcal{MN}_{n\times p}(\boldsymbol{\mu}, \mathbb{I}_n, \sigma^2\mathbb{I}_p), \tag{14}$$

which means that the lines of $\mathbf{X}$ are independent p-dimensional random vectors distributed as $\mathcal{N}_p(\mu_i, \sigma^2\mathbb{I}_p)$, for every line $i \in \{1, \dots, n\}$. Note that this model imposes an independence assumption between the features (columns) of $\mathbf{X}$.

To run a proper test, we need to control the selective type I error at level α. Ideally, we would like to define a p-value as follows:

$$p_{ideal}(\mathbf{x}) = \mathbb{P}_{\mathcal{H}_0^{\{C_1,C_2\}}}\big(T(\mathbf{X}) \geq T(\mathbf{x}) \mid C_1, C_2 \in C(\mathbf{X})\big),$$

with T being a test statistic. In [5], Gao *et al.* chose $T(\mathbf{X}) = ||\nu^\top \mathbf{X}||_2$, because its distribution under the null hypothesis can be straightforwardly derived. Indeed, noting that:

$$\nu^\top \mathbf{X} \sim \mathcal{MN}_{1\times p}\big(\nu^\top \mu,\ \nu\nu^\top,\ \sigma^2\mathbb{I}_p\big) = \mathcal{N}_p\big(\nu^\top \mu,\ \|\nu\|_2^2\sigma^2\mathbb{I}_p\big),$$

we have, under $\mathcal{H}_0^{\{C_1,C_2\}}$, that

$$||\nu^\top \mathbf{X}||_2 \sim ||\nu||_2 \sigma \chi_p. \tag{15}$$

However, p_{ideal} cannot be evaluated in practice as it depends on unknown parameters [5]. To address this issue, the authors in [5] propose to add technical events to the conditioning set, considering the following quantity:

$$p(\mathbf{x}) = \mathbb{P}_{\mathcal{H}_0^{\{C_1,C_2\}}}\Big(||\nu^T \mathbf{X}||_2 \geq ||\nu^T \mathbf{x}||_2 \mid C_1, C_2 \in \mathcal{C}(\mathbf{X}), \xi(\mathbf{X}, \mathbf{x})\Big), \tag{16}$$

as a p-value for (11), with $\xi(\mathbf{X}, \mathbf{x})$ denoting the additional events. To justify the choice of $\xi(\mathbf{X}, \mathbf{x})$, we follow the steps in [5]. First, we decompose the data matrix as follows:

$$\mathbf{X} = \pi_\nu^\perp \mathbf{X} + (\mathbb{I}_n - \pi_\nu^\perp)\mathbf{X}, \tag{17}$$

where $\pi_\nu^\perp$ is the orthogonal projection matrix onto the subspace orthogonal to ν, defined as:

$$\pi_\nu^\perp = \mathbb{I}_n - \frac{\nu\nu^\top}{||\nu||_2^2} \in \mathbb{R}^{n\times n}. \tag{18}$$

Then, we can write

$$\begin{aligned}
\mathbf{X} &= \pi_\nu^\perp \mathbf{X} + \frac{\nu\nu^T}{||\nu||_2^2}\mathbf{X} \\
&= \pi_\nu^\perp \mathbf{X} + \frac{\nu}{||\nu||_2^2}\frac{\nu^T\mathbf{X}}{||\nu^T\mathbf{X}||_2}||\nu^T\mathbf{X}||_2 \\
&= \pi_\nu^\perp \mathbf{X} + \frac{\nu}{||\nu||_2^2}\,\mathrm{dir}(\nu^T\mathbf{X})||\nu^T\mathbf{X}||_2,
\end{aligned} \tag{19}$$

where dir denotes the direction of a vector, that is,

$$\text{dir}(z) = \frac{z}{||z||_2}, z \in \mathbb{R}^p. \tag{20}$$

Since $\text{dir}(\nu^T \mathbf{X})$ and $\pi_\nu^\perp \mathbf{X}$ are unknown, we condition on their values for a realization $\mathbf{x}$ of $\mathbf{X}$, defining the additional events as:

$$\xi(\mathbf{X}, \mathbf{x}) = \big\{\pi_\nu^\perp \mathbf{X} = \pi_\nu^\perp \mathbf{x},\ \text{dir}(\nu^T \mathbf{X}) = \text{dir}(\nu^T \mathbf{x})\big\}.$$

Plugging the previous equation in (16) we obtain the following expression for the p-value:

$$\begin{aligned} p(x) = \mathbb{P}_{\mathcal{H}_0^{\{C_1,C_2\}}}\Big(||\nu^\top \mathbf{X}||_2 \geq ||\nu^\top x||_2 \,\Big|\, C_1, C_2 \in \mathcal{C}(\mathbf{X}), \\ \pi_\nu^\perp \mathbf{X} = \pi_\nu^\perp \mathbf{x},\ \text{dir}(\nu^\top \mathbf{X}) = \text{dir}(\nu^\top \mathbf{x})\Big). \end{aligned} \tag{21}$$

Now that we have defined the model, the hypothesis to be tested, and the corresponding p-value, we will show that this p-value can be computed under an analitically tractable form and that it properly controls the selective type I error. For this, we rely on the following technical lemma, whose proof is presented in Appendix A.

Lemma 1. *Under the null hypothesis, the following statements hold:*

$$\begin{aligned} &(i) \quad \|\nu^\top \mathbf{X}\|_2 \perp\!\!\!\perp \pi_\nu^\perp \mathbf{X}, \\ &(ii) \quad \|\nu^\top \mathbf{X}\|_2 \perp\!\!\!\perp \text{dir}(\nu^\top \mathbf{X}). \end{aligned}$$

Following from the previous result, we can rewrite the p-value as:

$$\begin{aligned} p(\mathbf{x}) = \mathbb{P}_{\mathcal{H}_0^{\{C_1,C_2\}}}\Big(||\nu^\top \mathbf{X}||_2 \geq ||\nu^\top \mathbf{x}||_2 \ | \\ C_1, C_2 \in \mathcal{C}(\pi_\nu^\perp \mathbf{X} + \frac{\nu}{||\nu||_2^2}\,\text{dir}(\nu^\top \mathbf{x})||\nu^\top \mathbf{x}||_2)\Big). \end{aligned} \tag{22}$$

Letting $\phi = ||\nu^\top \mathbf{X}||_2$ and $\mathbf{x}'(\phi) = \pi_\nu^\perp \mathbf{x} + \frac{\nu}{||\nu||_2^2}\,\text{dir}(\nu^\top \mathbf{x})\phi$, we can write

$$\begin{aligned} p(\mathbf{x} &= \mathbb{P}_{\mathcal{H}_0^{\{C_1,C_2\}}}\Big(\phi \geq ||\nu^\top \mathbf{x}||_2 \ | \ C_1, C_2 \in \mathcal{C}\big(\mathbf{x}'(\phi)\big)\Big) \\ &= \mathbb{P}_{\mathcal{H}_0^{\{C_1,C_2\}}}\Big(\phi \geq ||\nu^\top \mathbf{x}||_2 \ | \ \phi \in \mathcal{S}\big(\mathbf{x}; \{C_1, C_2\}\big)\Big), \end{aligned} \tag{23}$$

where

$$\mathcal{S}\big(\mathbf{x}, \{C_1, C_2\}\big) = \Big\{\phi \geq 0 : C_1, C_2 \in \mathcal{C}\big(\mathbf{x}'(\phi)\big)\Big\}. \tag{24}$$

As, under the null, we have that $\phi \sim ||\nu||_2 \sigma \chi_p$, we finally have:

$$p(\mathbf{x}) = 1 - \mathbb{F}_p\Big(\nu^\top \mathbf{x}; ||\nu||_2 \sigma; \mathcal{S}\big(\mathbf{x}; \{C_1, C_2\}\big)\Big), \tag{25}$$

where $\mathbb{F}_p(t; c; \mathcal{S})$ denotes the cumulative distribution function of a $c \cdot \chi_p$ random variable truncated to the set $\mathcal{S}$. This proves that the p-value (21) can be written in the analitically tractable form (25). This corresponds to the first statement in [5, Theorem 1], that we present below.

Theorem 1 (Theorem 1 in [5]). *For any non-overlapping groups of observations $C_1, C_2 \subseteq \{1, ..., n\}$ and any realization $\mathbf{x}$ of $\mathbf{X} \sim \mathcal{MN}_{n\times p}(\mu, \mathbb{I}_n, \sigma^2\mathbb{I}_p)$*

$$p(\mathbf{x}) = 1 - \mathbb{F}\big(\nu^\top \mathbf{x};\, ||\nu||_2\sigma;\, \mathcal{S}\big(\mathbf{x}; \{C_1, C_2\}\big)\big) \tag{26}$$

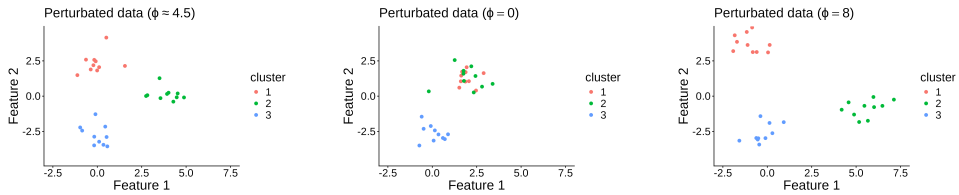
Furthermore, if $\mathcal{H}_0^{\{C_1,C_2\}}$ is true, then

$$\mathbb{P}_{\mathcal{H}_0^{\{C_1,C_2\}}}\Big(p(\mathbf{x}) \leq \alpha \mid C_1, C_2 \in \mathcal{C}(\mathbf{X})\Big) = \alpha, \quad \forall\, 0 \leq \alpha \leq 1. \tag{27}$$

The control of the selective type I error is proved in Appendix A. To shed some light on the truncation set $\mathcal{S}$, let us rewrite $\mathbf{x}'(\phi)$ as follows:

$$\begin{aligned}
\mathbf{x}'(\phi) &= \pi_\nu^\perp \mathbf{x} + \frac{\nu}{||\nu||_2}\operatorname{dir}(\nu^\top \mathbf{x})\phi \\
&= \mathbf{x} - \frac{\nu\nu^\top}{||\nu||_2^2}\mathbf{x} + \frac{\nu}{||\nu||_2}\operatorname{dir}(\nu^\top \mathbf{x})\phi \\
&= \mathbf{x} + \frac{\nu}{||\nu||_2^2}(\phi \operatorname{dir}(\nu^\top \mathbf{x}) - \nu^\top \mathbf{x}) \\
&= \mathbf{x} + \frac{\nu}{||\nu||_2^2}(\phi \operatorname{dir}(\nu^\top \mathbf{x}) - ||\nu^\top \mathbf{x}||_2 \operatorname{dir}(\nu^\top \mathbf{x})) \\
&= \mathbf{x} + \frac{\nu}{||\nu||_2^2}\operatorname{dir}(\nu^\top \mathbf{x})(\phi - ||\nu^\top \mathbf{x}||_2).
\end{aligned} \tag{28}$$

Consequently, $\mathbf{x}'(\phi)$ can be interpreted as a perturbed version of $\mathbf{x}$, where the compared clusters are either pushed appart or pulled together depending on the value of ϕ. When $\phi \leq ||\nu^T\mathbf{x}||$, the smaller ϕ is, the more the clusters are pulled together. If $\phi > ||\nu^T\mathbf{x}||$, the larger ϕ is, the more separate they are. More details are provided in Section 2.2 in [5]. Here, we illustrate these perturbations in Figure 3.



(a) (b) (c)

Figure 3: (a) Original dataset $\mathbf{x}'(\phi) = \mathbf{x}$ with $\phi = ||\nu^T\mathbf{x}|| \approx 4.5$, (b) perturbated dataset $\mathbf{x}'(\phi)$ with $\phi = 0$ and (c) a perturbated dataset $\mathbf{x}'(\phi)$ with $\phi = 8$. In all cases, C_1 and C_2 in $\nu(C_1, C_2)$ correspond to the red (1) and green (2) clusters.

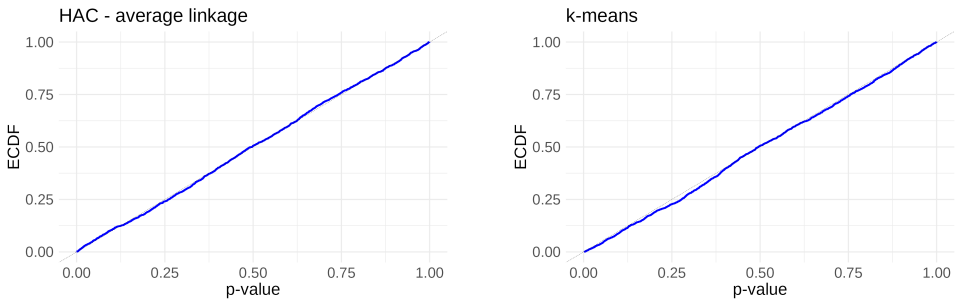
Thus, following from (24), $\mathcal{S}$ corresponds to the set of perturbations that preserve the clusters to be compared. The efficient determination of such set is required for the computation of (21). This can be carried out thanks to the strategies proposed by Gao *et al.* for several HAC algorithms (see [5, Section 3] for details).

3.2 Numerical experiments

3.2.1 Uniformly distributed p-values under the null hypothesis

Section 3.1 detailed the approach of Gao *et al.* for testing the equality of cluster means after implementing an HAC algorithm. In Theorem 1, we showed that the p-value defined in (21) controls the selective type I error for clustering.

To empirically illustrate this result, we simulated the null distribution of p-values, following the same strategy as in Section 2.2.3 but this time using (21) as a p-value for (11). For the simulations based on the HAC algorithm, we used the `clusterpval` library developed by Gao *et al.*, which implements the p-value (25). For the simulations based on the k-means algorithm, we used the `KmeansInference` library [3], which relies on a similar p-value adapted to the k-means procedure. The corresponding results are presented in Figure 4, which shows that the resulting p-values are indeed super-uniform under the null hypothesis.



(a) (b)

Figure 4: Empirical cumulative distribution functions of the p-values obtained after testing the equality of cluster means for (a) hierarchical agglomerative clustering (HAC) and (b) k-means algorithms. The empirical cumulative distribution functions (ECDF) were computed using $M = 2000$ samples drawn from a multivariate centered normal distribution, as detailed in Section 2.2.3.

3.2.2 Robustness to feature dependence

The control of the selective type I error for clustering relies on a strong assumption: the features must be independent (14). To assess whether the method is robust to this assumption, we repeated the strategy in the previous section and simulated the null distribution of p-values for $M = 2000$ realizations of:

$$\mathbf{X} \sim \mathcal{MN}_{n\times p}\big(\boldsymbol{\mu}, \mathbb{I}_n, \boldsymbol{\Sigma}\big), \tag{29}$$

with

$$\boldsymbol{\Sigma} = \rho^{|i-j|} \text{ for } 1 \leq i, j \leq p \text{ and } \rho \in (0, 1]. \tag{30}$$

Figure 5 shows that the p-values are no longer super-uniform under the null hypothesis, and therefore the selective type I error is no longer controlled when the independence assumption is not satisfied. Besides, we observe that the deviation from uniformity increases with the dependency parameter ρ. Indeed, the stronger the dependence between the features, the less the selective type I error is controlled. The deviation from super-uniformity also increases with the dimension, as illustrated in Figure 5 (c-d).

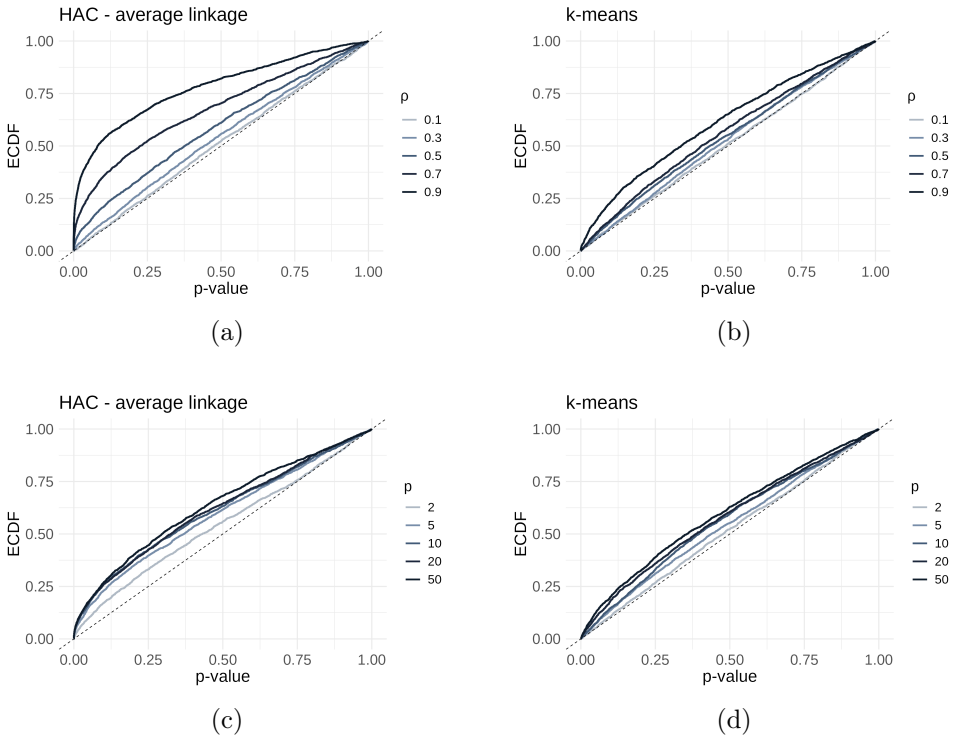


Figure 5: Empirical cumulative distribution functions (ECDF) of the p-values obtained after testing the equality of cluster means for hierarchical agglomerative clustering (HAC). The distribution functions were computed using $M = 2000$ samples drawn from (29) with correlation matrix (30) for (a-b) different values of ρ , with fixed $p = 5$, and (c-d) different values of p, for fixed $\rho = 0.5$.

3.3 Data applications

In this section, we assess the suitability of the method with real cases of application, where the data does not necessarily fit the model (14). In what follows, we first introduce the genotype dataset (1KGP cohort). We then analyze the output of the HAC clustering algorithm applied to this dataset. Finally, we assess whether the p-value proposed by Gao *et al.* provides statistical guarantees for this clustering.

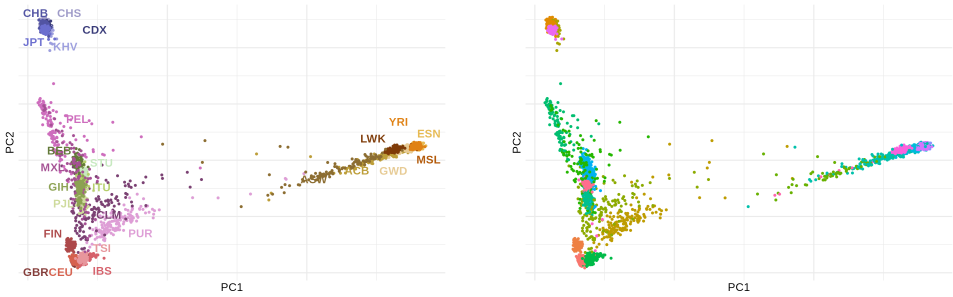
3.3.1 The 1KGP data set

The DNA sequence contains all the genetic information associated with an individual. It is composed of two complementary strands made of nucleotides, denoted by A, C, G, and T. Although these sequences are very long, the vast majority of the genome is shared among all individuals. Instead of relatively comparing DNA sequences, genetic databases are constructed by compar-

ing every individual's genome to the so-called *reference genome*. This yields a matrix of genetic variation, often referred to as *SNP matrix*, that serves as quantitative descriptor of the genetic variability of a human population. Here, we consider the SNP matrix of the 1000 Genomes Project (1KGP) data set, which contains genotype information of 3450 individuals from 26 human populations [1]. From this SNP matrix, a principal component analysis (PCA) can be performed in order to obtain the dataset `1KGP_100PC`, which we will use as a starting point for this analysis. Besides containing the genetic information of every individual, the 1KGP dataset also provides their ancestral population, that we detail in Appendix B.

3.3.2 Clustering

To try to capture the population structure of the 1KGP indivuduals, we perfomed HAC clustering and set the algorithm to choose K clusters, with K being the number of ancestral populations. Indeed, we consider the set of populations as an underlying true partition to which the clustering output will be compared. Figure 6 shows the 1KGP individuals represented on the first two principal components (PCs).



(a) (b)

Figure 6: Projection of the individuals onto the first two PCs, colored by population in (a) and by clusters obtained from HAC with Ward linkage on the first 100 PCs in (b).

The clusters obtained with the HAC algorithm in Figure 6(b) appear to be largely consistent with the population labels of the individuals. This can also be assessed by computing the confusion matrix comparing the clustering output and the population classification, as shown in Figure 7. Overall, the heatmap exhibits an approximately diagonal structure, indicating that each cluster mostly corresponds to a single population. However, some populations are spread across several clusters (PJL, ASW, GIH), and some clusters contain individuals from multiple populations (1, 3, 16).

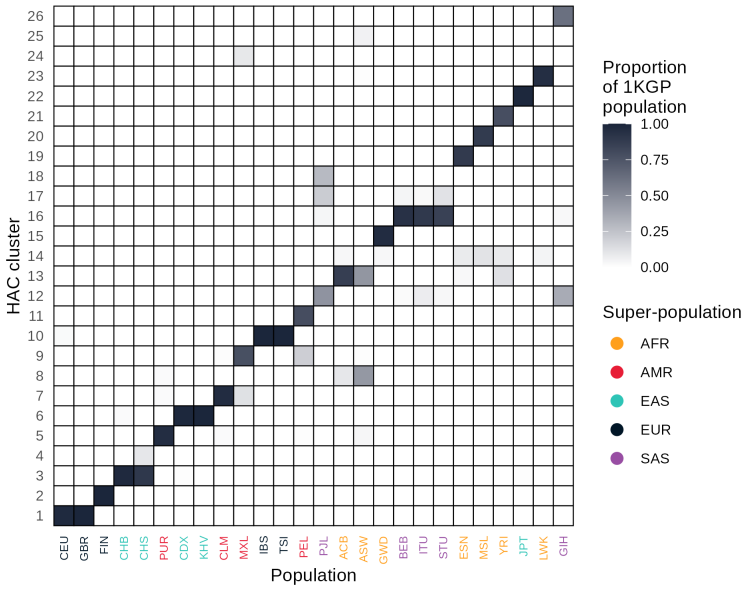


Figure 7: Confusion matrix comparing the clustering output and the population labels for the 1KGP individuals. Clusters were obtained using the HAC algorithm with Euclidean distance and the Ward linkage.

3.3.3 Suitability of Gao *et al.*'s approach

To assess whether the p-values of Gao *et al.* are consistent with the confusion matrix in Figure 7, we apply a post-clustering inference procedure to the clusters obtained via HAC. The p-values associated with comparisons between clusters containing individuals from the same super-population are expected to exceed a given significance threshold (typically 0.05), whereas those associated with clusters containing individuals from different super-populations should fall below this threshold.

So far, we have assumed that the data set fits model (14). However, with real data, the variance parameter σ is unknown. Gao *et al.* showed that [5, Theorem 4] by using an estimator that asymptotically overestimates the variance, such as:

$$\hat{\sigma}(\mathbf{x}) = \sqrt{\frac{1}{np-p}\sum_{i=1}^{n}\sum_{j=1}^{p}(x_{ij}-\bar{x}_j)^2}, \tag{31}$$

the selective type I error remains asymptotically controlled. Therefore, in this section we will be plugging (31) into (21), mantaining the statistical guarantees.

In Section 3.2.2, we showed that the p-value is sensitive to dimensionality. In practice, it is convenient to retain the maximum number of principal

components to minimize information loss. However, this comes at a cost to the statistical guarantees of Gao *et al.*'s method. To be statistically valid, it must control the selective type I error. However, a useful testing procedure should also be *powerful*. Panels (a–b) in Figure 8 show that, for two clusters composed of individuals from the same super-population, the p-value falls below 0.05 (the conventional significance level) when $p > 12$. This implies that the cluster means are considered significantly different, despite originating from the same super-population. Furthermore, panels (c–d) show that, for two clusters containing individuals from different but closely super-populations in the first two PCs, the dimension p must be greater than 6 for the p-value to drop below 0.05. Consequently, the p-value is both valid and powerful only for a small range of p, which substantially limits its practical applicability.

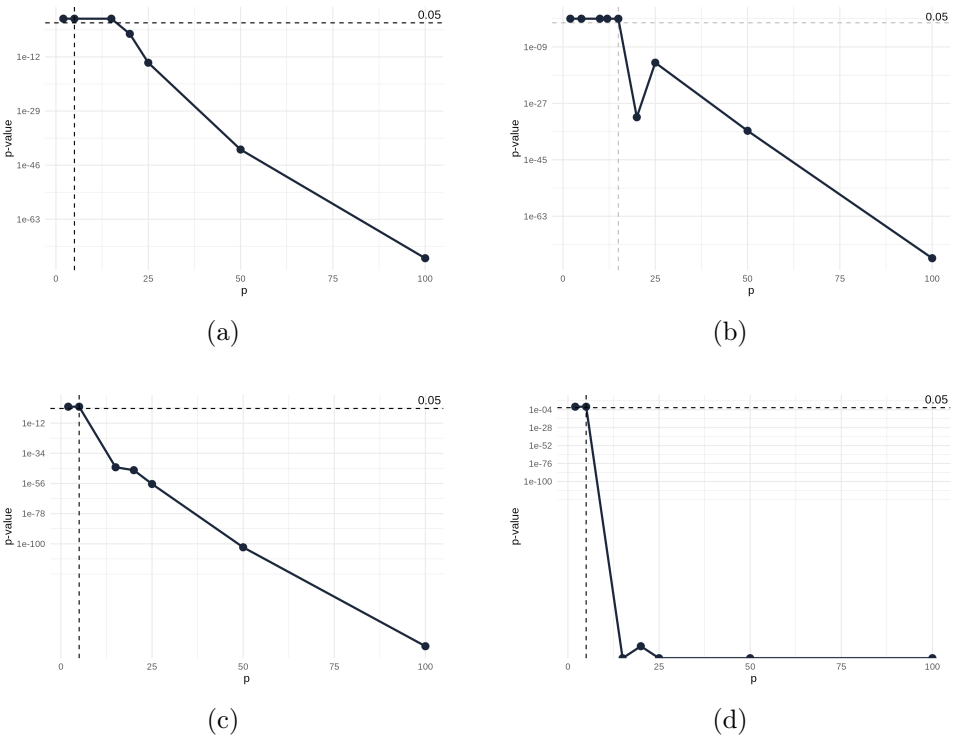


Figure 8: p-values (21) obtained from testing the equality of cluster means for clusters (a) $\{12, 26\}$, (b) $\{8, 13\}$, (c) $\{5, 7\}$ and (d) $\{9, 13\}$, as a function of the number of PCs used for clustering, that is, the dimension p of the feature space.

3.3.4 Assessing intercontinental vs intracontinental differences

Each population belongs to a super-population, corresponding to a larger grouping similar to a continent (see Table 1 in Appendix B). For exam-

ple, Mexican Ancestry in Los Angeles, California (MXL) and Puerto Rican in Puerto Rico (PUR) belong to the admixed-American (AMR) super-population, whereas Utah residents with Northern and Western European ancestry (CEU) belongs to the European (EUR) super-population. Intuitively, the p-values associated with tests between clusters containing individuals from different super-populations are expected to be close to 0, while the p-values associated with tests between clusters containing individuals from the same super-population are expected to be larger. To compute the p-values, we kept only the first 12 principal components following from the analyses presented in the previous section. The resulting p-values are shown in Figure 9. We observe that within each super-population, the p-values indicate that the clusters have equal means, as expected. Although a few p-values suggest that clusters containing individuals from different super-populations also have statistically equal means, the test appears to successfully recover the super-population structure of the data. Consequently, the method of Gao *et al.*'s seems to correctly detect broad population structure in this particular setting.

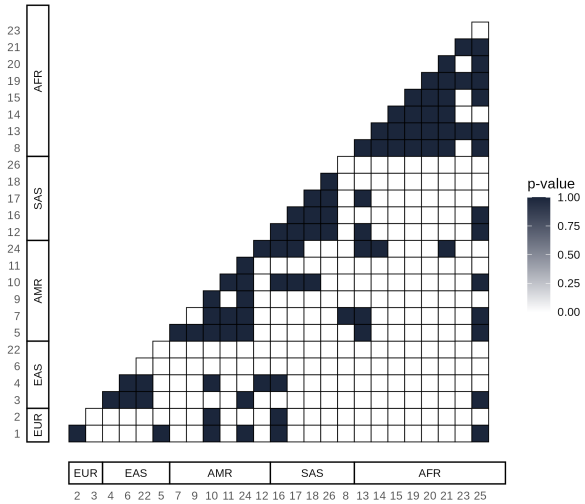


Figure 9: Heatmap of the p-values obtained using the `test_hier_clusters_exact` function applied to the first 12 PCs, with clusters obtained from HAC using Euclidean distance and the Ward.D linkage method on the first 100 principal components.

4 Discussion

Research on post-clustering inference is relatively recent, and the method of Gao *et al.* represents a seminal contribution within the conditional inference

framework. For the first time, they proposed a procedure to test whether there is a significant difference between the means of clusters estimated via HAC under the model (14). Importantly, their approach maintains asymptotic control of the selective type I error when the unknown parameter σ is over-estimated. Another key assumption of the model is that the features are independent, which is unlikely to hold in practice. In fact, we have shown that as features become more dependent, the p-value deviate further from super-uniformity. We also show that the robustness of the p-value in that setting decreases with the dimension p.

All of these limitations also appear when applying the method of Gao *et al.* to real data. Indeed, we have illustrated that it is not possible to simultaneously control the selective type I error and maintain high statistical power in practical applications. Nevertheless, the resulting p-values are able to detect broad class structures in some settings. To address this limitations, González-Delgado *et al.* [6] proposed a method that remains valid under arbitrary dependence structures between observations and features.

Code Availability

The code to reproduce the simulations and the real data analyses presented in this work can be found at https://github.com /JasminNeveu/projet_methodologique.

A Proofs

A.1 Proofs of Section 2

Proof of Proposition 1. Let G_X denote the generalized inverse (quantile function) of F_X, defined for $u \in [0, 1]$ as

$$G_X(u) = \inf\{x \in \mathbb{R} : F_X(x) \geq u\}. \tag{32}$$

By definition of the generalized inverse,

$$\{F_X(Y) \leq u\} = \{Y < G_X(u)\}. \tag{33}$$

Therefore,

$$\mathbb{P}\big(F_X(Y) \leq u\big) = \mathbb{P}\big(Y < G_X(u)\big) = F_Y\big(G_X(u)^-\big), \tag{34}$$

where $G_X(u)^-$ denotes the left limit at $G_X(u)$.

Since $X \preceq_{\text{st}} Y$, we have $F_Y \leq F_X$ pointwise, and thus

$$F_Y\big(G_X(u)^-\big) \leq F_X\big(G_X(u)^-\big). \tag{35}$$

By the defining property of the generalized inverse,

$$F_X\big(G_X(u)^-\big) \le u. \tag{36}$$

Combining these inequalities yields

$$\mathbb{P}\big(F_X(Y) \le u\big) \le u, \quad \forall\, u \in [0,1], \tag{37}$$

which proves that $F_X(Y)$ is super-uniform.

□

Proof of Remark 1. By the proposition 1, we have that $F_X(X) \sim \mathcal{SU}(0,1)$. If F_X is continuous, then $F_X(G_X(u)) = u$ for all $u \in [0,1]$, and hence

$$\mathbb{P}\big(F_X(X) \le u\big) = F_X\big(G_X(u)\big) = u, \quad \forall\, u \in [0,1], \tag{38}$$

which concludes the proof.

□

Proof of Proposition 2. By definition of the rejection rule,

$$\mathbb{P}_{\mathcal{H}_0}(\text{Reject } \mathcal{H}_0) = \mathbb{P}_{\mathcal{H}_0}(p \le \alpha). \tag{39}$$

Since p is a p-value for $\mathcal{H}_0$, it is super-uniform under $\mathcal{H}_0$, hence

$$\mathbb{P}_{\mathcal{H}_0}(p \le \alpha) \le \alpha. \tag{40}$$

This establishes control of the type I error at level α. □

Proof of Proposition 3. Let $F_{T'(X)}$ denote the distribution function of $T'(X)$ under $\mathcal{H}_0$. By Proposition 1, the stochastic dominance $T'(X) \preceq_{\text{st}} T(X)$ implies that

$$F_{T'(X)}\big(T(X)\big) \sim \mathcal{SU}(0,1).$$

By definition,

$$p(x) = \mathbb{P}_{\mathcal{H}_0}\big(T'(X) \ge T(x)\big) = 1 - F_{T'(X)}\big(T(x)\big). \tag{41}$$

Let $u \in [0,1]$. Then

$$\begin{aligned}
\mathbb{P}_{\mathcal{H}_0}\big(p(X) \le u\big) &= \mathbb{P}_{\mathcal{H}_0}\big(1 - F_{T'(X)}(T(X)) \le u\big) && (42)\\
&= \mathbb{P}_{\mathcal{H}_0}\big(F_{T'(X)}(T(X)) \ge 1-u\big) && (43)\\
&= 1 - \mathbb{P}_{\mathcal{H}_0}\big(F_{T'(X)}(T(X)) \le 1-u\big)\,. && (44)
\end{aligned}$$

Since $F_{T'(X)}(T(X))$ is super-uniform,

$$\mathbb{P}_{\mathcal{H}_0}\big(F_{T'(X)}(T(X)) \le 1-u\big) \le 1-u, \tag{45}$$

and therefore

$$\mathbb{P}_{\mathcal{H}_0}\big(p(X) \le u\big) \le u. \tag{46}$$

Thus, p is super-uniform under $\mathcal{H}_0$, and hence a p-value. □

Proof of Remark 2. When $T' = T$

$$p(X) = \mathbb{P}_{\mathcal{H}_0}\big(T(X') \geq T(X)\big) = 1 - F_T\big(T(X)\big), \tag{47}$$

By Remark 1, if F_T is continuous,

$$F_T\big(T(X)\big) \sim \mathcal{U}(0,1).$$

Consequently, for any $u \in [0,1]$,

$$\begin{aligned}
\mathbb{P}_{\mathcal{H}_0}\big(p(X) \leq u\big) &= \mathbb{P}_{\mathcal{H}_0}\big(1 - F_T(T(X)) \leq u\big) && (48)\\
&= \mathbb{P}_{\mathcal{H}_0}\big(F_T(T(X)) \geq 1 - u\big) && (49)\\
&= 1 - \mathbb{P}_{\mathcal{H}_0}\big(F_T(T(X)) \leq 1 - u\big) && (50)\\
&= 1 - (1 - u) && (51)\\
&= u. && (52)
\end{aligned}$$

Thus p is uniformly distributed on $[0,1]$ under $\mathcal{H}_0$. □

A.2 Proofs of Section 3

Proof of Lemma 1. i)

$$\begin{aligned}
\mathbb{E}(\nu^\top \mathbf{X}(\pi_\nu^\perp \mathbf{X})^T) &= \mathbb{E}(\nu^\top \mathbf{X}\mathbf{X}^\top (\pi_\nu^\perp)^\top)\\
&= \nu^\top \mathbb{E}(\mathbf{X}\mathbf{X}^\top)(\pi_\nu^\perp)^\top\\
&= \nu^\top (\mathbb{V}(\mathbf{X}) + \mathbb{E}(\mathbf{X})\mathbb{E}(\mathbf{X})^\top)(\pi_\nu^\perp)^\top.
\end{aligned} \tag{53}$$

Let $X \sim \mathcal{MN}(\mu, \mathbf{U}, \mathbf{\Sigma})$. Then, let us state $\mathbb{V}(\mathbf{X}) = \mathbf{U}\,\mathrm{Trace}(\mathbf{\Sigma})$. We prove the previous equality as a technical lemma after the proof.

In this case $\mathbb{V}(\mathbf{X}) = \mathbb{I}_n\,\mathrm{Trace}(\sigma^2 \mathbb{I}_p) = \sigma^2 p \mathbb{I}_n$. Therefore,

$$\begin{aligned}
\mathbb{E}(\nu^\top \mathbf{X}(\pi_\nu^\perp \mathbf{X})^T) &= \nu^\top (\sigma^2 p \mathbb{I}_n + \mathbb{E}(\mathbf{X})\mathbb{E}(\mathbf{X})^\top)(\pi_\nu^\perp)^\top\\
&= \sigma^2 p \nu^\top (\pi_\nu^\perp)^\top + \nu^\top \mathbb{E}(\mathbf{X})\mathbb{E}(\mathbf{X})^\top (\pi_\nu^\perp)^\top\\
&= \sigma^2 p \nu^\top (\mathbb{I}_n - \frac{\nu\nu^\top}{||\nu||_2^2}) + \mathbb{E}(\nu^\top \mathbf{X})\mathbb{E}(\pi_\nu^\perp \mathbf{X})^\top\\
&= \sigma^2 p (\nu^\top - \frac{||\nu||_2^2 \nu^\top}{||\nu||_2^2}) + \mathbb{E}(\nu^\top \mathbf{X})\mathbb{E}((\pi_\nu^\perp \mathbf{X})^\top)\\
&= \mathbb{E}(\nu^\top \mathbf{X})\mathbb{E}((\pi_\nu^\perp \mathbf{X})^\top).
\end{aligned} \tag{54}$$

ii)

$$\nu^T \mathbf{X} \sim \mathcal{MN}_{1\times p}(\nu^\top \mu, \nu\nu^\top, \sigma^2 \mathbb{I}_p) = \mathcal{N}_p(\nu^\top \mu, ||\nu||_2^2 \sigma^2 \mathbb{I}_p). \tag{55}$$

So $\nu^T \mathbf{X} \sim \mathcal{N}_p(0, ||\nu||_2^2 \sigma^2 \mathbb{I}_p)$ under $\mathcal{H}_0$. Yet, the density of a centered Gaussian random vector depends only on its norm, due to symmetry with respect to the origin. So $||\nu^\top \mathbf{X}||$ and $\mathrm{dir}(\nu^\top \mathbf{X})$ are independent.

□

Lemma 2. *Let $X \sim \mathcal{MN}(\mu, \mathbf{U}, \mathbf{\Sigma})$, then $\mathbb{V}(\mathbf{X}) = \mathbf{U}\,\mathrm{Trace}(\mathbf{\Sigma})$.*

Proof of Lemma 2.

$$\begin{aligned} \text{Let } \mathbf{Y} = \mathbf{X} - \mu \\ \mathbb{V}(\mathbf{X}) &= \mathbb{E}[(\mathbf{X}-\mu)(\mathbf{X}-\mu)^\top] \\ &= \mathbb{E}(\mathbf{Y}\mathbf{Y}^\top). \end{aligned} \tag{56}$$

And

$$\begin{aligned} (\mathbf{Y}\mathbf{Y}^T)_{ij} &= \sum_{k=1}^{p} \mathbf{Y}_{ik}\mathbf{Y}_{kj}^\top \\ &= \sum_{k=1}^{p} \mathbf{Y}_{ik}\mathbf{Y}_{jk}. \end{aligned} \tag{57}$$

Therefore,

$$\mathbb{E}((\mathbf{Y}\mathbf{Y}^\top)_{ij}) = \sum_{k=1}^{p} \mathbb{E}(\mathbf{Y}_{ik}\mathbf{Y}_{jk}). \tag{58}$$

As $\mathbf{Y}$ is centered, by definition of a normal matrix we have:

$$\begin{aligned} \mathbb{E}((\mathbf{Y}\mathbf{Y}^\top)_{ij}) &= \sum_{k=1}^{p} \mathbf{U}_{ij}\Sigma_{kk} \\ &= \mathbf{U}_{ij}\sum_{k=1}^{p}\Sigma_{kk} \\ &= \mathbf{U}_{ij}\,\mathrm{Trace}(\Sigma). \end{aligned} \tag{59}$$

□

Proof of Equation 27 in Theorem 1. From the definition (21), we have that for all $\mathbf{x} \in \mathbb{R}^{n\times p}$

$$\mathbb{P}_{\mathcal{H}_0^{\{C_1,C_2\}}}\big(p(\mathbf{x}) \le \alpha \mid C_1, C_2 \in \mathcal{C}(\mathbf{X}), \pi_\nu^\perp \mathbf{X} = \pi_\nu^\perp \mathbf{x}, \mathrm{dir}(\nu^\top \mathbf{X}) = \mathrm{dir}(\nu^\top \mathbf{x})\big) = \alpha. \tag{60}$$

Therefore, we have

$$\begin{aligned} &\mathbb{P}_{\mathcal{H}_0^{\{C_1,C_2\}}}\big(p(\mathbf{x}) \le \alpha \,\big|\, C_1, C_2 \in \mathcal{C}(\mathbf{X})\big) \\ &= \mathbb{E}_{\mathcal{H}_0^{\{C_1,C_2\}}}\Big[\mathbb{1}\{p(\mathbf{x}) \le \alpha\} \,\Big|\, C_1, C_2 \in \mathcal{C}(\mathbf{X})\Big] \\ &= \mathbb{E}_{\mathcal{H}_0^{\{C_1,C_2\}}}\Big[\mathbb{E}_{\mathcal{H}_0^{\{C_1,C_2\}}}\Big[\mathbb{1}\{p(\mathbf{x}) \le \alpha\} \,\Big|\, C_1, C_2 \in \mathcal{C}(\mathbf{X}), \pi_\nu^\perp \mathbf{X} = \pi_\nu^\perp \mathbf{x}, \\ &\qquad\qquad \mathrm{dir}(\nu^\top \mathbf{X}) = \mathrm{dir}(\nu^\top \mathbf{x})\Big] \,\Big|\, C_1, C_2 \in \mathcal{C}(\mathbf{X})\Big] \\ &= \mathbb{E}_{\mathcal{H}_0^{\{C_1,C_2\}}}\big[\alpha \,\big|\, C_1, C_2 \in \mathcal{C}(\mathbf{X})\big] \\ &= \alpha, \end{aligned} \tag{61}$$

where the second equality follows from the law of total expectation. □

B Data Description

Table 1: Description of the different populations of individuals.
EAS = East Asian, SAS = South Asian, AFR = African, EUR = European, AMR = admixed American.

Description	Population	Superpopulation
Chinese Dai in Xishuangbanna, China	CDX	EAS
Han Chinese in Beijing, China	CHB	EAS
Japanese in Tokyo, Japan	JPT	EAS
Kinh in Ho Chi Minh City, Vietnam	KHV	EAS
Southern Han Chinese, China	CHS	EAS
Bengali in Bangladesh	BEB	SAS
Gujarati Indian in Houston, Texas	GIH	SAS
Indian Telugu in the UK	ITU	SAS
Punjabi in Lahore, Pakistan	PJL	SAS
Sri Lankan Tamil in the UK	STU	SAS
African Ancestry in Southwest US	ASW	AFR
African Caribbean in Barbados	ACB	AFR
Esan in Nigeria	ESN	AFR
Gambian in Western Division, The Gambia	GWD	AFR
Luhya in Webuye, Kenya	LWK	AFR
Mende in Sierra Leone	MSL	AFR
Yoruba in Ibadan, Nigeria	YRI	AFR
British in England and Scotland	GBR	EUR
Finnish in Finland	FIN	EUR
Iberian populations in Spain	IBS	EUR
Toscani in Italy	TSI	EUR
Utah residents with Northern and Western European ancestry	CEU	EUR
Colombian in Medellín, Colombia	CLM	AMR
Mexican Ancestry in Los Angeles, California	MXL	AMR
Peruvian in Lima, Peru	PEL	AMR
Puerto Rican in Puerto Rico	PUR	AMR

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